

Connecting the Dots in AMR Surveillance

Revealing resistance structure and anticipating what comes next

Giordano De Marzo¹, Alfredo De Bellis², Chiara Mastrogioseppe³

¹Center for Data and Methods, University of Konstanz, Konstanz, Germany

²Center for Health Emergencies, Fondazione Bruno Kessler, Trento, Italy

³Department of Biomedical Sciences, Humanitas University, Pieve Emanuele, Italy

Abstract

Antibiotic resistance is usually studied one species–antibiotic pair at a time, yet resistance mechanisms often affect several drugs and can be acquired together [1, 2], so the pairwise view tracks how much resistance there is but not how it is organised across species and drugs. Using the Pfizer ATLAS surveillance programme distributed through the Vivli AMR Register (394 species, 43 antibiotics, 2004–2024) [3], we score each species–drug pair not by absolute resistance but by whether the species is more resistant than expected from its own overall resistance and the drug’s overall failure rate; these comparative-resistance links form a bipartite species–antibiotic network. The network splits into coherent organism–drug groups, and within each group comparative resistance is strongly nested, with broadly resistant species defeating essentially the drugs that narrower species defeat plus harder ones. Applying the Fitness–Complexity framework from economic networks [4], we rank species by the breadth and difficulty of the drugs they defeat and antibiotics by how demanding they are to defeat; although no clinical labels enter the construction, antibiotic complexity rises across the WHO Access, Watch and Reserve tiers and ESKAPEE species receive the highest fitness [5, 6]. Finally, projecting the network onto its antibiotic side, a parameter-free exposure score anticipates new comparative-resistance links with an AUC of 0.71 and remains informative up to five years ahead. We release the framework as an interactive atlas (<https://giordano-demarzo.github.io/resistome-atlas/>) with full reproducibility code [7].

Objectives

We pursued three linked objectives: (i) turn routine counts into a pathogen–antibiotic network that isolates pair-specific comparative resistance from the general resistance level of a species and the general vulnerability of a drug; (ii) test whether comparative-resistance profiles are nested, as partially ordered accumulation of mechanisms predicts [8, 9], and use Fitness–Complexity to rank every species and drug within the resulting groups; and (iii) test prospectively whether a species’ position relative to empirically similar drugs anticipates which drug it defeats next.

Methods

From surveillance counts to a pathogen–antibiotic network

We consider as a starting point the annual species–antibiotic counts from the Pfizer ATLAS surveillance program distributed through the Vivli AMR Register, spanning 394 species, 43 antibiotics and 2004–2024 [3]. We define the resistance fraction r_{pa} of pathogen p against antibiotic a as $r_{pa} = n_{pa}^R / N_{pa}$, where n_{pa}^R is the number of times p as been tested to be resistant, while N_{pa} is the total number of p isolates tested with antibiotic a . We only considered pairs with $N_{pa} \geq 30$ and we denote the set of sufficiently observed pairs by $\mathcal{O}$. Similar results are obtained for $N_{pa} \geq 10, 50$.

Absolute resistance may reflect either broadly resistant species, broadly ineffective antibiotics, or an unusually resistant species and antibiotic pair. While important, the first two phenomena can be easily tracked with standard surveillance methods. Here instead we focus on the last case, so a situation in which a pathogen shows resistance to an antibiotic that is larger than what we could expect considering its general resistance against the other antibiotics and the average effectiveness of the drug itself against the other pathogens. We can then introduce the revealed comparative resistance of species p against antibiotic a as

$$\text{RCA}_{pa} = \frac{r_{pa}}{R_p} \frac{T}{A_a}$$

Here R_p is the average resistance of pathogen p across all antibiotics, A_a is the average resistance of pathogens against antibiotic a and T the average resistance of all pathogens against all antibiotics. For instance, $\text{RCA}_{pa} = 2$ indicates that pathogen p resistance fraction is twice the value expected from the species and antibiotics averages. This measure is invariant to proportional increases across the entire matrix and therefore isolates comparative rather than overall resistance.

Finally, we introduce the binary matrix $M_{pa} = \mathbf{1}\{\text{RCA}_{pa} \geq 1\}$ of comparative resistance (from now on just resistance for shortness) by only considering pathogens–antibiotics pairs with a comparative resistance larger than one. This allows to filter out trivial signals such as pathogens resistant to almost everything or old antibiotics defeated nowadays by virtually

all pathogens. Note that for insufficiently observed pairs $M_{pa} = 0$; thus, while ones imply resistance, zeros do not imply confirmed susceptibility, and RCA should not be interpreted as a clinical breakpoint. The matrix M_{pa} can be visualized as a bipartite network (Figure 1a). We can group all pathogens on one side, all the antibiotics on the other, and then we link each pathogen p to all the antibiotics a for which $M_{pa} = 1$. i.e. links go from pathogens to the antibiotics they resist to with a comparative resistance larger than one.

Nested resistance profiles

Looking at the antibiotic resistance from the network point of view allows to understand patterns and features that would otherwise be invisible by simply looking at individual pathogens-drugs pairs. This is similar to what happens in epidemic spreading, where an individual can act as a super-spreader, but such a feature can only be identified by looking at the whole social contact network, not at the individual level characteristics. The first aspect we investigate is whether our bipartite network structure allows to reconstruct well known groups of pathogens. Indeed, intrinsic barriers, resistance mechanisms, and testing practices differ across taxa: for example, the Gram-negative outer membrane limits the entry of many large antibiotics, while susceptibility interpretation rules also vary across organisms [10, 11]. Here the idea is that very different pathogens (for instance Gram-positive vs Gram-negative bacteria) will be susceptible to a very different set of drugs, and this should be reflected in our network. To do so, we apply a spectral clustering algorithm [12] to the binary matrix M_{pa} , allowing us to obtain a meaningful set of pathogens-antibiotics communities.

Within each group not all pathogens are equivalent. Some show a comparative resistance against most drugs, others are linked to only a few. More precisely, we need to test whether the pathogens that resist most drugs, typically also resist the antibiotics resisted by the less resistant pathogens. This is a so called nested structure and it is a ubiquitous characteristic of many ecological systems. It suggests that pathogens accumulate capabilities or features that make them antibiotic-resistant, from the most simple, that only guarantee resistance against basic antibiotics, to the most sophisticated, that make them insusceptible even to advanced drugs [8, 9]. These capabilities could be for instance target modification, reduced membrane permeability or efflux pumps. Every time a pathogen, through mutations, acquires one of these capabilities, it adds to its repertoire resistance to a new set of drugs.

Fitness of pathogens and complexity of drugs

When networks are nested, nodes within them can be ranked on the basis of how many capabilities they acquired and their diversification. In our framework, pathogens resisting many drugs should be assigned a high rank, while the opposite should be true for antibiotics, i.e. a drug resisted by many should be assigned a low rank. The Economic Fitness and Complexity (EFC) algorithm allows to quantify these aspects in a data driven way, starting from the network structure only [4]. It assigns to each pathogen p a fitness F_p and to each antibiotic a a complexity Q_a (Figure 1b). Fitness is a quantification of the number and sophistication of a pathogen’s capabilities, while complexity is a measure of how difficult it is to defeat, in terms of capabilities possessed, a drug. Since pathogens and drugs are connected in the network structure and do not live in an isolated space, the EFC algorithm combines information from both pathogens and antibiotics to generate meaningful ranking. The fitness F_p of a species p reflects both the breadth of its comparative resistance profile and the difficulty of the antibiotics it defeats. Conversely, the complexity Q_a of an antibiotic a depends not simply on how rarely a drug is defeated, but on which species defeat it. A drug has high complexity only when it is defeated exclusively by species with broad comparative resistance profiles. Even one species with a narrow profile substantially lowers its complexity, since this would imply that limited capabilities are needed to defeat the antibiotic. Fitness and complexity are therefore estimated jointly, with each helping to determine the other. Starting from $F_p^{(0)} = Q_a^{(0)} = 1$, the pipeline iterates

$$\tilde{F}_p^{(n)} = \sum_a M_{pa} Q_a^{(n-1)} \quad , \quad \tilde{Q}_a^{(n)} = \left(\sum_p \frac{M_{pa}}{F_p^{(n-1)}} \right)^{-1}$$

normalizing F and Q by their means at each step. Because heterogeneous non-nested structures can produce collapsed rankings, scores were computed separately within each group and standardized before pooling. Note that Fitness is a network score, unrelated to evolutionary fitness, while complexity is not drug potency or clinical efficacy.

Two external classifications were used for validation only after scoring: the WHO Access, Watch and Reserve stewardship tiers [5] for antibiotic complexity, and the ESKAPEE group, which extends the ESKAPE mnemonic to include *Escherichia coli* [6], for pathogen fitness. Neither entered the network, grouping or scores.

The antibiotic similarity network and resistance forecasting

Since capabilities accumulate and resistance emerges from this stacking process, the set of drugs a pathogen is expected to resist in the future, is not a random sample of what it does not defeat today. A pathogen may have almost all capabilities needed to resist a drug, only missing a small mutation to finally defeat it. These aspects can be quantified by building a similarity network of drugs, where drugs are connected when at least one pathogen defeats both of them (Figure 1c).

This may indeed be an indication of the two antibiotics requiring similar capabilities to be defeated, even if it does not identify the cause [2, 13]. In particular, we define the antibiotic-antibiotic network in terms of the similarity matrix $W_{ab}^A = \sum_p \theta(M_{pa}^{(t)} M_{pb}^{(t)})$, where a, b are any pair of antibiotics and $\theta(x) = 1$ when $x > 0$.

As a last step, we can use the antibiotic network to forecast future resistance. The idea is that resistance does not spread at random on the antibiotic network: a pathogen tends to become resistant to drugs that are similar and thus connected to those it already resists (Figure 1d). For every sufficiently observed pair (p, a) with no link at a given year t , we can then define the exposure of drug a with respect to pathogen p as the share of drug a 's neighbourhood already defeated by species p :

$$\rho_{pa} = \frac{\sum_o W_{ao}^A M_{po}^{(t)}}{\sum_o W_{ao}^A}$$

The exposure ρ_{pa} is high when pathogen p already resists most of the drugs that are similar to a , while it's low otherwise. Therefore, we expect the probability for a pathogen to develop future resistance to be positively correlated with the corresponding exposure.

Results

The network splits into coherent groups with nested resistance

Pooling 2015–2024 gave 1,932 sufficiently observed species–antibiotic cells in a 139-by-40 bipartite network, of which 991 carried a comparative-resistance link ($M_{pa} = 1$). We first asked whether this network, built from surveillance counts alone, reproduces known organism–drug domains. Spectral clustering of M partitioned species and antibiotics into five groups that, although no taxonomic, drug-class or stewardship label entered the construction, matched recognizable clinical ecosystems: one large Gram-negative group (Enterobacterales together with *Acinetobacter*, *Pseudomonas* and *Stenotrophomonas*) organised around β -lactams, cephalosporins and carbapenems; a Gram-positive group of staphylococci and streptococci built on macrolides, fluoroquinolones and oxacillin; a distinct enterococcal group defined by the glycopeptides and daptomycin (the vancomycin-resistant enterococcus ecosystem); and two anaerobe groups centred on *Clostridia* (metronidazole, penicillin) and *Bacteroides* (clindamycin, cefoxitin, tigecycline). Each group therefore pairs a coherent set of organisms with the drugs represented in its testing profile, exactly as differing intrinsic barriers and susceptibility rules would predict [10, 11].

We then asked whether, *within* these groups, comparative-resistance profiles are nested: whether the species that resist many drugs also resist largely the same drugs defeated by species with narrower profiles. Nestedness was strong within groups (a z -score of 10.51 against a degree-preserving null): narrower profiles were contained in broader ones, and the drugs defeated by the fewest species were reached only by species that defeated almost everything else in their group. Measured on M as a whole, by contrast, the same statistic was negligible and even below the null ($z = -4.30$). Comparative resistance is therefore modular: a *Klebsiella* and an *Enterococcus* do not share one biologically interpretable ordering, yet within each group resistance profiles are strongly ordered, consistent with resistance capabilities being acquired in a partially ordered sequence [8, 9].

Fitness and complexity rank species and drugs within each group

Having established that resistance is nested within groups, we ranked the species and drugs of each group with the Fitness–Complexity algorithm, applied separately within every group and standardized before pooling. The iteration converged to a stable fitness F_p for every species and complexity Q_a for every drug. We then compared these rankings against biological and clinical classifications that had been withheld from the construction. Antibiotic complexity increased across the WHO Access, Watch and Reserve tiers (Spearman $\rho = 0.5895$, $p = 9.84 \times 10^{-5}$, 38 drugs; $\rho = 0.6244$, $p = 8.49 \times 10^{-4}$ within the largest group), so the purely topological ranking recovered the stewardship gradient [5] (Figure 2a). The species ranking behaved analogously: ESKAPEE taxa [6] received the highest fitness, with median standardised fitness 1.044 across 16 matching taxa versus -0.387 for the other 107 (one-sided Mann–Whitney $p = 4.60 \times 10^{-6}$; Figure 2b). Because neither classification entered the network, its grouping or the scores, these agreements are external consistency checks rather than clinical validation: both rankings use only the topology of comparative-resistance links.

Comparative resistance expands toward similar drugs, and the next link can be anticipated

The bipartite network provides more than a static ranking of species and antibiotics: its structure also describes how comparative-resistance profiles grow. Projecting the network onto its antibiotic side yields the similarity network W^A , in which two drugs are connected when at least one species defeats both, grouping drugs by shared resistance behaviour rather than by chemical class (Figure 3a). Resistance expands locally on this network: the comparative-resistance profile of *Klebsiella pneumoniae* grew from 12 linked drugs in 2014 to 17 in 2019 and 21 in 2024, with new links appearing next to drugs it already defeated while distant regions stayed empty (Figure 3b–d). This is a descriptive account of how profiles expand; it does not establish molecular acquisition or transmission, but it motivates using network proximity to

anticipate which drug a species defeats next.

As detailed in the methods, we can use exposure score ρ_{pa} to forecast resistance, and evaluated it in a rolling-origin design that rebuilds the network and the scores at each origin. Of 1,839 eligible pathogen–drug pair-years, 131 (7.12%) crossed the comparative-resistance threshold in the following year. Exposure separated these events sharply: 51 of 307 pairs (16.61%) in the top exposure sixth crossed within one year, against 14 of 307 (4.56%) in the bottom sixth, a 3.6-fold difference (Figure 4a). Ranking future links by exposure reached a pooled AUC of 0.7076 (origin-specific 0.6688–0.7703), against 0.5123 for drug prevalence and 0.6835 for the pair’s current comparative resistance, and it exceeded all 200 node-label permutations ($p < 0.005$; Figure 4b). An AUC of 0.7076 means that a future event outranked a non-event 70.76% of the time; it measures ranking, not individual probability. Notably, the signal persisted beyond one year, with similar results holding from one- to five-year horizons. The findings are also stable to the isolate cutoff: rerunning the pipeline at $n_{\min} = 10, 30, 50$ preserved all results.

Impact of the work

What this adds to surveillance. A surveillance report says how much resistance there is, not where it is heading. From the same counts, our framework adds three complementary views: a network *ranking* of every species and drug within its group; an antibiotic *similarity network* built from shared resistance behaviour rather than chemical class; and a *ranked watch-list* of species–drug pairs lying close to existing comparative-resistance links, with the best-connected “contagion hub” drugs acting as early-warning sentinels when testing capacity is limited.

Interpretable by construction. The whole chain is short and inspectable (counts $\rightarrow r_{pa} \rightarrow \text{RCA}_{pa} \rightarrow M \rightarrow W^A \rightarrow \rho_{pa}$): nothing is trained, fitted or tuned. A pair ranks highly simply because specific neighbouring drugs are already defeated by that species, a rationale a microbiologist can check against the isolate counts, and the analysis re-runs as soon as new counts arrive.

The Resistome Atlas. The interactive dashboard (<https://giordano-demarzo.github.io/resistome-atlas/>) ships with a deterministic end-to-end script and output tables. For any pathogen it shows its comparative-resistance profile, its rank within its group alongside WHO AWaRe labels, and the drugs it is closest to defeating, letting surveillance teams, stewardship groups and laboratories prioritise pairs for confirmatory testing. Dashboard probabilities come from a separate one-variable calibration of exposure and are not clinical guidance.

Deployment and scope. The pipeline needs only isolate counts by species, drug and year. Because comparative resistance is defined relative to the analysed population, each country, region or time window has its own network; temporal re-estimation is demonstrated here, while geographical re-estimation (the present table pools countries) and testing the inferred relationships against genomic resistance determinants [14] are natural next steps.

Limits. Comparative resistance is relative, so a pair can be flagged at modest prevalence or stay unremarkable despite severe absolute resistance; ATLAS is a surveillance sample rather than a random one [15]; and a zero in M merges below-threshold pairs with untested ones, so the network partly reflects testing coverage. The number of groups is fixed at five, AWaRe and ESKAPEE provide external consistency rather than clinical validation, AUC measures ranking rather than individual risk, and nothing here establishes causation, transmission or treatment failure.

Tables/figures

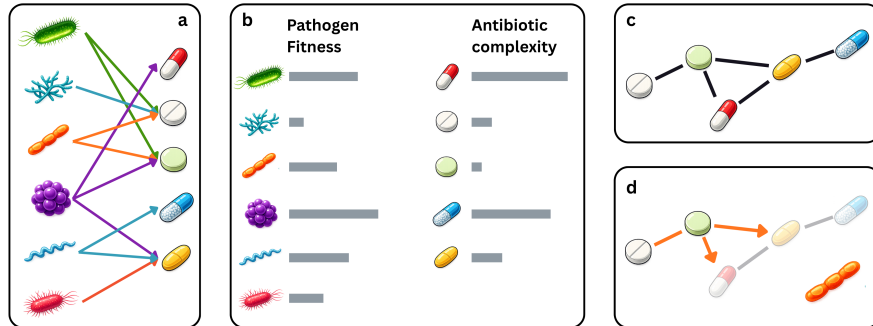


Figure 1: Overview of the framework. From surveillance counts, the pipeline builds a comparative resistance network and uses its structure to rank organisms and drugs and to forecast new resistance. (a) Comparative-resistance links define a bipartite network: each pathogen (left) is connected to the antibiotics (right) it resists more than expected from its own resistance level and the drug’s average failure rate ($M_{pa} = 1$). (b) The Economic Fitness–Complexity algorithm ranks pathogens by the breadth and difficulty of the drugs they defeat (fitness F_p) and antibiotics by how demanding they are to defeat (complexity Q_a); bar length indicates the score. (c) Projecting the bipartite network onto its antibiotic side yields the similarity network W^A , connecting drugs defeated by the same species. (d) Resistance expands locally on this network: a species already resistant to a drug’s neighbours (orange) is the likeliest to defeat it next, which is quantified by the exposure score ρ_{pa} . Organism and drug icons are illustrative.

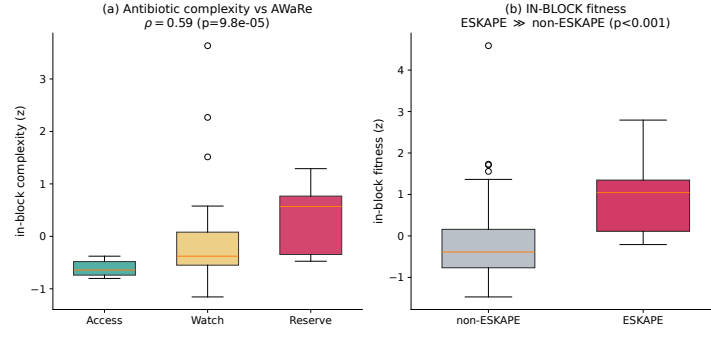


Figure 2: Network rankings recover known clinical and biological groups. Species fitness increases when a species is comparatively resistant to many antibiotics, especially to drugs resisted by few other species; antibiotic complexity is higher when a drug is comparatively resisted only by high-fitness species. Both scores are calculated from the network alone, without using external labels. (a) Antibiotic complexity increases from the WHO Access to Watch and Reserve groups. (b) ESKAPEE species receive the highest fitness within their groups. Fitness is a network score, not evolutionary fitness, and complexity does not measure drug potency.

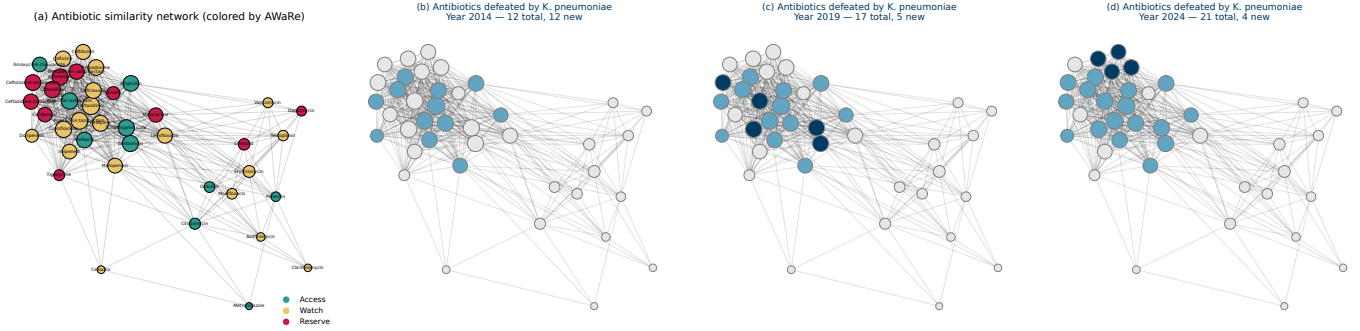


Figure 3: Antibiotic similarity network and expansion of resistance in *Klebsiella pneumoniae*. (a) Antibiotics (nodes) are connected when they are comparatively resisted by the same species: grey lines (links) show these similarities, while node colours indicate the WHO Access, Watch and Reserve classes for illustration only. (b–d) Blue nodes mark antibiotics with a comparative-resistance link to *K. pneumoniae* in 2014, 2019 and 2024. Over time, new links tend to appear near antibiotics that were already linked, suggesting that comparative resistance expands locally across the antibiotic similarity network. Light blue nodes mark links already present in the previous snapshot, whereas dark blue nodes mark links newly observed since the previous snapshot. The predicted event is the formation of a new link with $RCA \geq 1$, not demonstrated biological acquisition or clinical resistance.

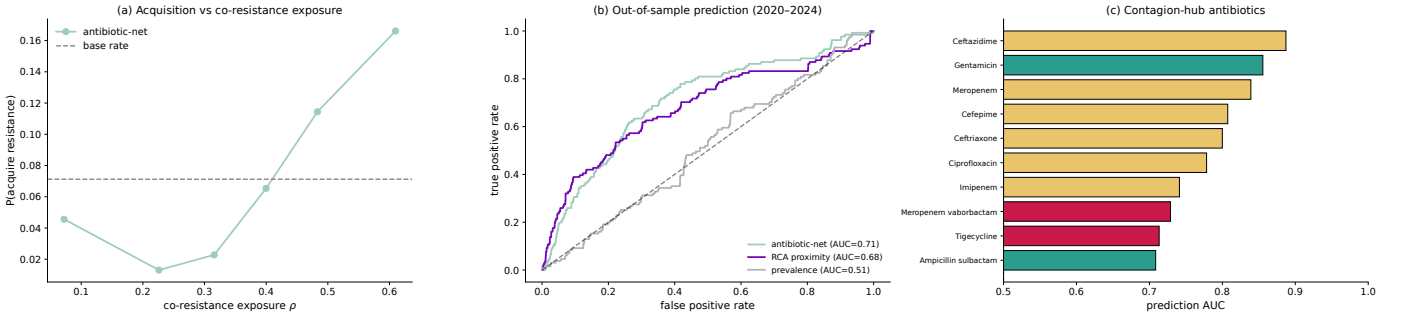


Figure 4: Forecasting the next drug. Network exposure anticipates new comparative-resistance links. For each pathogen–antibiotic pair that is not yet linked in year t , exposure measures how strongly that pathogen is already linked to antibiotics similar to the candidate drug. (a) The plot shows the observed fraction of pairs that develop a new comparative-resistance link in the following year: this fraction (y-axis) is higher when exposure is high (x-axis). (b) Exposure ranks future links better than the pair’s current RCA and than how commonly the drug is already linked across pathogens (AUC 0.71 vs 0.68 and 0.51). (c) Drug-specific AUCs identify the antibiotics whose neighbourhoods in the antibiotic similarity network are most informative for predicting which pathogens will develop a new comparative-resistance link to that drug in the following year.

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